

Problem

For the circuit in Fig. 7.100,

$$v = 90e^{-50t} \text{ V}$$

and

$$i = 30e^{-50t} \text{ A}, \quad t > 0$$

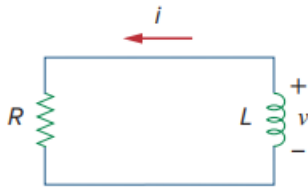
(a) Find L and R .

(b) Determine the time constant.

(c) Calculate the initial energy in the inductor.

(d) What fraction of the initial energy is dissipated in 10 ms?

Fig. 7.100:



Step-by-step solution

Step 1 of 5

(a)

Refer to Figure 7.100 in the text book.

Apply Kirchhoff's voltage law in the circuit.

$$Ri - v = 0$$

$$v = Ri$$

Substitute $90e^{-50t} \text{ V}$ for v and $30e^{-50t} \text{ A}$ for i .

$$90 = R(30)$$

$$R = \frac{90}{30}$$

$$R = 3 \, \Omega$$

Thus, the value of the resistor is $\boxed{3 \, \Omega}$.

Step 2 of 5

Use Ohm's law to write the expression of the voltage across the inductor.

$$v = -L \frac{di}{dt}$$

Substitute $90e^{-50t}$ V for v and $30e^{-50t}$ A for i .

$$90e^{-50t} = -L \frac{d}{dt}(30e^{-50t})$$

$$90e^{-50t} = -30L(-50)e^{-50t}$$

$$90 = 1500L$$

$$L = 60 \text{ mH}$$

Therefore, the value of the inductor is 60 mH.

Step 3 of 5

(b)

Write the expression for the time constant of the RL circuit.

$$\tau = \frac{L}{R}$$

Substitute 60 mH for L and 3Ω for R .

$$\tau = \frac{60 \times 10^{-3}}{3}$$

$$= 20 \times 10^{-3}$$

$$= 20 \text{ ms}$$

Thus, the time-constant of the circuit is 20 ms.

Step 4 of 5

(c)

Determine the initial value for the current passing through the inductor.

$$i_L(0) = 30e^{-50(0)}$$
$$= 30 \text{ A}$$

Determine the initial energy stored in the inductor.

$$E_L(0) = \frac{1}{2} Li^2(0)$$
$$= \frac{1}{2} (60 \times 10^{-3}) (30)^2$$
$$= 27 \text{ J}$$

Thus, the initial energy stored in the inductor is 27 J.

Step 5 of 5

(d)

Determine the energy stored in the inductor at $t = 10 \text{ ms}$.

$$\begin{aligned}E_L(10 \text{ ms}) &= \frac{1}{2} L i^2(10 \text{ ms}) \\&= \frac{1}{2} (60 \times 10^{-3}) \left(30 e^{-50(10 \times 10^{-3})} \right)^2 \\&= 27 \left(e^{-50(10 \times 10^{-3})} \right)^2 \\&= 9.933 \text{ J}\end{aligned}$$

The fraction of energy dissipated in first 10 ms is,

$$\begin{aligned}\Delta E &= \frac{E_L(0) - E_L(10 \text{ ms})}{E_L(0)} \times 100 \\&= \frac{27 - 9.933}{27} \times 100 \\&= 63.2\%\end{aligned}$$

Thus, the fraction of energy dissipated in first 10 ms is 63.2%.